

Advancing Healthcare with Multimodal AI: Enhanced Diagnosis, Cognitive Monitoring, and Inclusive Medical Communication

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Abstract. This doctoral research explores multimodal deep learning to enhance diagnosis, cognitive monitoring, and communication accessibility in healthcare. Across three clinical domains - cardiovascular diagnostics, neurocognitive assessment, and sign language recognition - we develop transformer-based pipelines tailored for real-time applicability, interpretability, and privacy compliance.

In the cardiovascular setting, we propose spatio-temporal models for non-invasive estimation of FFR and iFR from single-view coronary angiography, integrating convolutional and transformer encoders with multi-task supervision. These innovations culminate in the STARFLOW system, a deployable platform achieving high accuracy in multicenter studies, and protected by national and EU patents.

For cognitive screening, we introduce pipelines that analyze facial micro-movements and eye-gaze trajectories using phase-based motion magnification, attention-based encoders, and a novel Time-Aware Positional Embedding (TAPE) module. These systems enable automated scoring of MMSE and MoCA using commodity tablets, facilitating contactless and scalable cognitive health tools.

To promote inclusivity, we release MultiMedaLIS, the first multimodal dataset of Italian medical sign language, and develop a GDPR-compliant radar-based recognition pipeline using Range-Doppler Maps and temporal transformers. This system ensures robust gesture classification under privacy and environmental constraints.

Collectively, this work contributes a unified framework for interpretable and adaptive AI in healthcare, advancing real-world integration through methodological innovation, validated deployment, and alignment with clinical needs.

1 Research Problem and Motivation

Modern healthcare systems face increasing pressure to deliver accurate, accessible, and equitable diagnostic and monitoring services - especially in resource-limited, privacy-sensitive, or real-time critical contexts. Despite recent advances in AI for medical imaging and human behavior analysis, most existing solutions

remain limited by stringent data requirements, poor generalizability across domains, lack of interpretability, and minimal attention to deployment constraints such as latency, energy efficiency, and patient privacy.

My research addresses this gap by investigating how multimodal, transformer-based deep learning can be harnessed to support three core challenges in clinical intelligence: (i) non-invasive physiological assessment of coronary artery disease, (ii) contactless cognitive and behavioral monitoring, and (iii) inclusive medical communication through sign language recognition. These domains share critical constraints - temporal complexity, sparse supervision, and privacy sensitivity - that demand rethinking current AI pipelines beyond conventional fully supervised, single-modal approaches.

The central problem I tackle is how to design spatio-temporal deep learning architectures that are not only accurate, but also clinically interpretable, deployable on edge devices, and adaptive to distributional shifts. This involves leveraging transformer architectures, prompt-based adaptation, radar imaging, and federated strategies to meet real-world clinical requirements.

By grounding these innovations in multimodal datasets, real hospital workflows, and open challenges in accessibility, this work aligns closely with the MIC-CAI community’s mission to advance medical image computing and computer-assisted interventions with tangible clinical impact.

2 Background

Modern approaches in each of the three target domains - cardiovascular diagnostics, cognitive assessment, and medical sign language recognition - have achieved notable progress, yet several limitations persist that hinder real-world and clinical adoption.

Cardiovascular physiology estimation. Recent years have seen a surge in deep learning applications for non-invasive cardiovascular, particularly for estimating Fractional Flow Reserve (FFR) from coronary imaging [4]. Convolutional architectures have been applied to static angiographic frames, but these often ignore contrast flow dynamics, require manual vessel segmentation, and struggle to generalize across centers. Commercial solutions like *Heartflow Analysis* by HeartFlow Inc. and *Medis QFR* by Medis Medical Imaging Systems B.V. offer good diagnostic accuracy, yet depend on CT acquisition or dual-view angiography and require computationally expensive fluid simulations, limiting their scalability and real-time usability. Spatio-temporal modeling of full angiography sequences has only recently been explored. Some models incorporate temporal reasoning [13], but most still assume multi-view inputs, omit clinical metadata, or lack interpretability - key factors for real-world adoption. Transformer-based architectures, proven effective in natural image understanding, are now being adapted to medical sequences [2] and multi-modal inputs [22], yet their integration into end-to-end, interpretable, and single-view frameworks for coronary physiology prediction remains limited.

Cognitive and behavioral monitoring. Traditional cognitive screening instruments, such as the *Mini-Mental State Examination (MMSE)* and the *Montreal Cognitive Assessment (MoCA)*, are time-intensive, require administration by trained clinicians, and remain susceptible to subjectivity and inter-rater variability. To overcome these limitations, recent research has explored the use of eye-tracking [25] and facial dynamics [26] as unobtrusive digital biomarkers. Early work in this domain has focused on handcrafted features extracted from facial landmarks [1] or gaze trajectories [23], often requiring expensive equipment and constrained lab settings. More recent approaches begin to explore multimodal pipelines [24], but these often neglect the inherent temporal variability in human behavior, fail to predict standardized cognitive scores directly, and are rarely optimized for mobile or real-time use. End-to-end systems capable of fusing spatiotemporal visual inputs with structured behavioral metadata remain scarce. As a result, current methods fall short of enabling scalable, objective, and accessible cognitive assessment in real-world environments.

Sign language recognition for medical communication. Most existing SLR systems rely on RGB [8,27], depth [10], or skeletal pose data [9], which are vulnerable to lighting variations, occlusions, and pose significant privacy concerns - especially in clinical environments. Wearable sensor-based solutions [11] hinder natural communication, and the development of domain-specific vocabularies for medical settings remains largely underexplored. While radar-based gesture recognition has recently gained traction for general-purpose applications [12], it has not been applied to the healthcare domain or adapted to the linguistic and contextual nuances of Italian Sign Language (LIS). Furthermore, multimodal fusion strategies involving radar remain underutilized, and until our MultiMedaLIS initiative [15], no public datasets existed for radar sign recognition or for medical sign language recognition - limiting the development and benchmarking of inclusive AI communication tools for privacy-constrained environments or for clinical use.

While prior work has made strides within cardiovascular physiology estimation, cognitive assessment, and medical sign-language recognition respectively, it still lacks: (i) single-view, temporally aware and clinically interpretable models for coronary physiology; (ii) mobile, privacy-preserving multimodal pipelines that directly infer standardized cognitive scores; and (iii) inclusive, radar augmented LIS frameworks and benchmark datasets that enable robust communication in resource-constrained healthcare settings.

3 Scientific Approach

The scientific approach of my doctoral research is grounded in the design of multimodal, interpretable, and clinically deployable deep learning systems. The work spans three domains of healthcare - cardiovascular diagnostics, cognitive and behavioral monitoring, and accessible medical communication - each characterized by unique challenges in terms of data availability, clinical feasibility, and privacy. To address these, I combined spatio-temporal deep learning, transformer-based

architectures, and domain-specific preprocessing pipelines, with the overarching aim of enabling accurate, robust, and real-time AI systems for clinical practice.

AI for Cardiovascular Diagnostics. The methodological foundation in this domain is the non-invasive estimation of Fractional Flow Reserve (FFR) and Instantaneous Wave-Free Ratio (iFR) from standard coronary angiography (CA). Unlike invasive catheter-based measurements, our models learn to capture contrast-agent dynamics and vessel morphology directly from image sequences. In [19], we proposed a dual-view transformer architecture that integrates two angiographic projections through global self-attention. This design explicitly models vessel structure and contrast propagation across views and frames, yielding both continuous FFR/iFR predictions and binary outputs. Results confirmed that temporal attention can approximate invasive physiology, establishing a foundation for data-driven surrogates. To improve scalability, [18] introduced a single-view convolutional transformer model with factorized axial attention. Here, spatio-temporal dependencies are decomposed: spatial attention encodes intra-frame vessel geometry, temporal attention captures flow across frames, and convolutional front-ends provide low-level feature extraction. Multi-task heads jointly perform regression and classification around thresholds (FFR=0.80, iFR=0.89). This design eliminates the need for dual-view acquisition, improving efficiency and compatibility with routine workflows. The STARFLOW system [7] consolidates these advances into a deployable end-to-end platform. STARFLOW uses a 3D ResNet-18 encoder followed by a transformer decoder that attends to hemodynamically relevant regions, integrating clinical metadata (age, LVEF, comorbidities, ACS/CCS status) through learned embeddings. Outputs include binary classification and regression for both FFR and iFR. Standardized preprocessing ensures robustness across acquisition protocols. In a multicenter study (N=389), STARFLOW achieved 87.3% accuracy with sensitivity 82.4% and specificity 92.2%, validating readiness for clinical integration. The approach has led to Italian and European patent filings [5,6], underlining its translational potential.

AI-Driven Cognitive and Behavioral Monitoring. The second research axis focuses on unobtrusive, objective, and scalable cognitive assessment, moving beyond paper-based screening. The methodology leverages fine-grained facial micro-movements and oculomotor dynamics as digital biomarkers of cognitive decline. In [20], we developed a pipeline for tablet-based video analysis during cognitive tasks. Preprocessing includes stabilization, facial region extraction (MTCNN), and Phase-Based Motion Magnification (PBMM) to amplify subtle micro-expressions otherwise invisible to the human eye. Motion History Images (MHIs) then summarize temporal activity into compact representations. The second stage fuses MHIs and structured metadata (task identifiers, performance scores, fatigue markers) using a dual-branch neural network: a MobileViT-based visual encoder with trainable 3D convolutional downsampling, and a structured branch for metadata. Outputs are concatenated and regressed to Mini-Mental State Examination (MMSE) scores. Experiments show that motion magnification significantly improves both accuracy and F1, highlighting the relevance of

involuntary facial dynamics as biomarkers. In [16], we extended this paradigm to eye-gaze trajectories captured with consumer tablets equipped with gaming eye trackers. Participants completed seven gamified cognitive tasks mapped to MoCA subdomains. A key innovation is the Time-Aware Positional Embedding (TAPE) module, which encodes irregular gaze sequences into temporally consistent tokens, mitigating variability across users and devices. These enriched tokens are processed by a transformer encoder with regression heads for both global and subdomain MoCA scores. Achieving up to 89.9% balanced accuracy, this system avoids handcrafted features and supports contactless, real-time cognitive monitoring. Optimized for mobile deployment, it enables scalable, longitudinal screening in clinics and home environments. Together, these contributions establish a methodological framework for AI-driven neurocognitive assessment using only commodity hardware, paving the way for remote, continuous monitoring.

Sign Language Recognition for Medical Communication. The third domain addresses inclusivity by enabling sign language communication in clinical contexts. We developed the MultiMedaLIS dataset [15,3], the first corpus tailored to medical Italian Sign Language (LIS). It contains over 25,000 isolated signs across 126 classes, with synchronized mm-wave radar, RGB-D, IR, and stereoscopic video recordings. Data were captured using portable sensors (Infineon XENSIV 60GHz radar, Intel Realsense D455, Kinect v1, ZED 2), and annotated through a validated protocol with native LIS signers. A foot-operated interface ensured natural signing and precise annotation, guaranteeing both linguistic and clinical fidelity. Building on this resource, [14] introduced the first radar-only SLR system for healthcare. Range-Doppler Maps (13 FPS, 3 antennas) are processed with a residual autoencoder to compress motion patterns into latent embeddings, followed by a transformer decoder for temporal classification. Radar-based sensing offers GDPR compliance by avoiding facial imagery, and ensures robustness to lighting, clothing, and environmental geometry. Preliminary results on MultiMedaLIS confirm high accuracy and real-time performance, validating the feasibility of deploying SLR at the patient’s bedside.

4 Proposed Solution

The core contribution of my doctoral research lies in the development of novel deep learning systems that translate multimodal signals into clinically actionable insights while addressing critical constraints such as data scarcity, privacy, and real-time deployment. These solutions span three healthcare domains and are unified by a shared emphasis on temporal modeling, interpretable predictions, and deployability in real-world settings.

Cardiovascular Physiology Estimation from Angiography. In the context of coronary artery disease, I designed a progression of deep architectures that allow non-invasive prediction of FFR/iFR indices from X-ray angiography - circumventing the need for pressure wires or vasodilators. The initial dual-view model [19] exploits self-attention to jointly capture anatomical and flow-related

information from synchronized projections, offering a first proof-of-concept for attention-based physiology estimation. To address scalability and routine deployment, I introduced a more efficient single-view model [18] that factorizes spatio-temporal attention into separate branches and incorporates a multi-task head to improve robustness near clinical cutoffs. This shift enables compatibility with standard acquisition protocols while reducing computational overhead. The complete solution is embodied in the STARFLOW system [7], which translates these models into an end-to-end clinical tool. STARFLOW integrates convolutional feature extractors, transformer-based sequence modeling, and patient metadata embeddings into a unified diagnostic interface. Its contributions include real-time inference, per-frame uncertainty estimates, and visual explanations to support physician trust. The system has been validated in a multicenter cohort and patented at both national and international levels [5,6], marking a significant step toward regulatory approval and clinical integration.

Contactless Cognitive Screening via Behavioral Biomarkers. In response to the limitations of traditional cognitive assessments, I proposed two contactless pipelines for evaluating neurocognitive function through visual and oculomotor signals. The first solution [20] targets facial micro-movements captured during tablet-based exercises. A motion magnification module amplifies subtle motor anomalies, while temporal features are extracted using Motion History Images and processed by a lightweight dual-branch architecture combining visual and structured task metadata. The model accurately regresses MMSE scores, establishing facial expressivity as a viable digital biomarker. Complementing this, I developed an eye-tracking-based system for MoCA score estimation [16]. The novelty lies in the TAPE (Time-Aware Positional Encoding) module, which allows transformers to process irregularly sampled gaze sequences by interpolating learned temporal embeddings. This enables robust regression of cognitive subdomain scores from gaze-driven visual tokens, without handcrafted features. The entire system runs on consumer tablets with integrated gaze sensors, offering an accessible solution for real-time, remote cognitive screening. Both approaches support continuous monitoring in aging populations and reduce reliance on expert-administered tests.

Radar-Based Sign Language Recognition for Inclusive Communication. To address communication barriers in healthcare, I developed the first radar-based sign language recognition pipeline tailored to medical settings. The MultiMedaLIS dataset [15,3] serves as the foundation, offering a diverse corpus of LIS signs captured through RGB-D, stereo, and mm-wave radar. My proposed solution [14] uses only Range-Doppler Maps to preserve privacy while maintaining gesture fidelity. A residual autoencoder compresses high-dimensional radar sequences into gesture embeddings, which are then decoded via a temporal transformer for classification. Unlike vision-based systems, this radar-centric design is invariant to lighting, appearance, and background geometry, ensuring deployment feasibility in uncontrolled environments such as emergency rooms or wards. It also satisfies GDPR requirements by eliminating identifiable imagery. Preliminary evaluations on MultiMedaLIS demonstrate strong classification accuracy,

and the system offers a scalable foundation for multimodal extensions and personalization.

Across all domains, my solutions introduce novel architectures (e.g., axial transformers, TAPE, radar autoencoders), modular design pipelines, and real-world validation strategies. Together, these contributions advance the state of the art in multimodal, explainable, and clinically ready AI for health.

5 Results and Contribution

My doctoral research has produced concrete contributions across three interconnected clinical domains, each validated through peer-reviewed publications, quantitative evaluation, and - in some cases - clinical deployment.

Cardiovascular Diagnostics. The STARFLOW pipeline [7], integrating convolutional and transformer components for spatio-temporal modeling of coronary angiography, has been validated in a multicenter clinical study with 389 patients. It achieved an overall classification accuracy of 87.3%, with sensitivity and specificity of 82.4% and 92.2%, respectively - matching or exceeding current invasive standards without requiring pressure wires. These results significantly outperform prior non-invasive methods. The system’s interpretability, robustness to acquisition variability, and real-time performance support its adoption in clinical workflows. The innovation has led to both an Italian patent [5] and a European extension [6], confirming its translational value.

Cognitive and Behavioral Monitoring. In the cognitive domain, the MMSE prediction model using magnified facial micro-movements [20] achieved an F1-score improvement of over 5% compared to non-magnified baselines, demonstrating the clinical relevance of imperceptible motor signals. The MoCA estimation pipeline [16], based on eye-gaze trajectories processed via the novel TAPE module, reached balanced accuracy scores up to 89.9%, surpassing standard positional encoding strategies. These results validate the potential of contactless, sensor-based cognitive monitoring systems that scale to mobile platforms and support telemedicine deployment.

Medical Sign Language Recognition. The MultiMedaLIS dataset [15,3] introduced a new multimodal benchmark for Italian medical sign language recognition, including 25’000+ annotated signs across 126 classes and four sensor modalities. On this dataset, the radar-only SLR system [14] demonstrated strong gesture classification performance while ensuring GDPR compliance. Its architecture, combining residual autoencoding and temporal transformers, achieves real-time inference and robustness to environmental variability. These results establish a novel privacy-preserving modality for clinical communication support.

Summary of Contributions. I introduced novel architectures for spatio-temporal learning, including axial transformer decompositions for coronary angiography and the TAPE module for eye-gaze encoding. My work led to end-to-end AI systems, such as STARFLOW and a radar-based sign language recognition pipeline, both validated in real-world clinical settings. I contributed new multimodal datasets, like MultiMedaLIS, supporting research on inclusive and

privacy-aware AI in medicine. Finally, I demonstrated clinical translation through deployment-ready tools, cross-center validations, and two patent filings supporting future regulatory integration.

Collectively, these results advance the state of the art in multimodal, interpretable, and deployable AI for healthcare by targeting real clinical gaps in diagnosis, monitoring, and communication.

6 Open Challenges and Future Work

Despite promising results, several challenges remain that define the roadmap ahead. **Generalization Across Clinical Settings:** Although STARFLOW and related models were validated in multicenter cohorts, variability in acquisition protocols and populations may limit robustness. Future work will address domain generalization via adversarial alignment, test-time adaptation, and uncertainty aware calibration. **Interpretable Multimodal Fusion:** Combining facial video, gaze, radar, and metadata requires principled fusion strategies. We will explore cross-modal transformers and attention mechanisms that preserve modality-specific cues while enabling interpretable decisions. **Scalable Cognitive Monitoring:** For deployment in homes and community clinics, pipelines must be hardware-standardized, user-friendly, support adaptive calibration, and integrate with electronic health records for longitudinal use. **Continual Learning and Online Adaptation:** Radar-based SLR must adapt to new users, signs, and scenarios. Future work targets rehearsal-free continual learning, prompt-based domain expansion, and online fine-tuning to avoid forgetting while retaining real-time performance. **Federated and Efficient Learning:** Privacy-preserving scalability will be pursued through post-training quantization (HC-PTQ [17]) for edge deployment, data-free continual learning (MAPT), and federated frameworks such as FedDETR [21] for angiography detection and Fed-CTVTCL-CLIP for visuo-textual adaptation without data sharing. **Clinical Trials and Regulatory Translation:** STARFLOW and the gaze-based MoCA predictor are entering translational phases, requiring multicenter trials, cost-effectiveness studies, and regulatory approval (e.g., CE marking, FDA pre-submission) in collaboration with hospitals.

6.1 Long-Term Goals

My long-term research vision is to advance **privacy-preserving, trustworthy, and adaptive AI systems** for healthcare and other safety-critical domains. Beyond demonstrating state-of-the-art accuracy, I aim to develop models that are interpretable, generalizable across institutions, and optimized for deployment on edge devices. This involves synergistic use of post-training quantization techniques such as HC-PTQ [17], privacy-aware federated learning frameworks like FedDETR [21] for coronary angiography and Fed-CTVTCL-CLIP for multimodal prompt adaptation, and data-free continual learning strategies to ensure robustness without data retention.

Looking ahead, I plan to extend the methodological framework established in my dissertation to broader domains, including neurology (e.g., Parkinson’s motor assessment), geriatrics (frailty screening), and tele-rehabilitation. At the same time, I will pursue translational validation through clinical trials, usability studies, and regulatory pathways, ensuring that algorithmic innovations become deployable medical tools. My overarching goal is to bridge the gap between academic progress and societal impact, ensuring that AI technologies not only advance research frontiers but also tangibly improve healthcare delivery, accessibility, and sustainability.

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References

1. AbdAlmageed, W., Mirzaalian, H., Guo, X., Randolph, L.M., et al.: Assessment of facial morphologic features in patients with congenital adrenal hyperplasia using deep learning. *JAMA network open* **3**(11), e2022199 (2020)
2. Akan, T., Alp, S., Bhuiyan, M.S., Helmy, T., Orr, A.W., et al.: Viviechoformer: deep video regressor predicting ejection fraction. *Journal of imaging informatics in medicine* pp. 1–12 (2024)
3. Caligiore, G., Mineo, R., Spampinato, C., Ragonese, E., Palazzo, S., Fontana, S.: Multisource approaches to italian sign language (LIS) recognition: Insights from the multimedalis dataset. In: Dell’Orletta, F., Lenci, A., Montemagni, S., Sprugnoli, R. (eds.) *Proceedings of the 10th Italian Conference on Computational Linguistics (CLiC-it 2024)*. pp. 132–140. *CEUR Workshop Proceedings*, Pisa, Italy (Dec 2024), <https://aclanthology.org/2024.clicit-1.17/>
4. Cho, H., Lee, J.G., Kang, S.J., Kim, W.J., Choi, S.Y., other: Angiography-based machine learning for predicting fractional flow reserve in intermediate coronary artery lesions. *Journal of the American Heart Association* **8**(4), e011685 (2019)
5. De Ferrari, G.M., D’Ascenzo, F., Aldinucci, M., Hughes, J.M., Millesimo, M., De Filippo, O., Iannaccone, M., Grosso Marra, W., Quadri, G., Proietto Salanitri, F., Bellitto, G., Spampinato, C., Mineo, R., Palazzo, S.: Metodo di diagnosi/prognosi real-time mediante ai in ambito cardiologico (2024), patent office: Ufficio italiano brevetti e marchi (UIBM)
6. De Ferrari, G.M., D’Ascenzo, F., Aldinucci, M., Hughes, J.M., Millesimo, M., De Filippo, O., Iannaccone, M., Grosso Marra, W., Quadri, G., Proietto Salanitri, F., Bellitto, G., Spampinato, C., Mineo, R., Palazzo, S.: Angiographic image

- processing method and system (2025), patent office: European Union Intellectual Property Office (EUIPO)
7. De Filippo, O., Mineo, R., Millesimo, M., Wańha, W., Proietto Salanitri, F., Greco, A., Leone, A.M., Franchin, L., Palazzo, S., Quadri, G., et al.: Non-invasive physiological assessment of intermediate coronary stenoses from plain angiography through artificial intelligence: the starflow system. *European Heart Journal-Quality of Care and Clinical Outcomes* **11**(3), 343–352 (2025)
 8. Huang, J., Zhou, W., Zhang, Q., Li, H., Li, W.: Video-based sign language recognition without temporal segmentation. In: *Proceedings of the AAAI conference on artificial intelligence*. vol. 32 (2018)
 9. Jiang, S., Sun, B., Wang, L., Bai, Y., Li, K., Fu, Y.: Skeleton aware multi-modal sign language recognition. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 3413–3423 (2021)
 10. Jing, L., Vahdani, E., Huenerfauth, M., Tian, Y.: Recognizing american sign language manual signs from rgb-d videos. *arXiv preprint arXiv:1906.02851* (2019)
 11. Kudrinko, K., Flavin, E., Zhu, X., Li, Q.: Wearable sensor-based sign language recognition: A comprehensive review. *IEEE Reviews in Biomedical Engineering* **14**, 82–97 (2020)
 12. Lien, J., Gillian, N., Karagozler, M.E., Amihoud, P., Schwesig, C., Olson, E., Raja, H., Poupyrev, I.: Soli: Ubiquitous gesture sensing with millimeter wave radar. *ACM Transactions on Graphics (TOG)* **35**(4), 1–19 (2016)
 13. Liu, P., Liu, G.: A two-stage coronary artery stenosis detection method. In: *Proceedings of the 2023 4th International Symposium on Artificial Intelligence for Medicine Science*. pp. 265–269 (2023)
 14. Mineo, R., Caligiore, G., Proietto Salanitri, F., Kavasidis, I., Polikovsky, S., Fontana, S., Ragonese, E., Spampinato, C., Palazzo, S.: Radar-based imaging for sign language recognition in medical communication. In: *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer (2025)
 15. Mineo, R., Caligiore, G., Spampinato, C., Fontana, S., Palazzo, S., Ragonese, E.: Sign language recognition for patient-doctor communication: A multimedia/multimodal dataset. In: *2024 IEEE 8th Forum on Research and Technologies for Society and Industry Innovation (RTSI)*. pp. 202–207. IEEE (2024)
 16. Mineo, R., Kavasidis, I., Salanitri, F.P., Passarello, L., Piccininno, G., Masciale, V., Anselmo, A., Convertino, C., Rotondi, D., Laurieri, N., et al.: Automated moca score estimation using eye-gaze data and vision transformers. In: *2025 IEEE 38th International Symposium on Computer-Based Medical Systems (CBMS)*. pp. 963–968. IEEE (2025)
 17. Mineo, R., Palazzo, S., Spampinato, C., Rundo, F.: Hc-ptq: Poincaré-based hyperbolic clustering for data-free quantization of vision transformers. In: *2025 IEEE/CVF International Conference on Computer Vision Workshops* (2025)
 18. Mineo, R., Salanitri, F.P., Bellitto, G., Kavasidis, I., De Filippo, O., Millesimo, M., De Ferrari, G., Aldinucci, M., Giordano, D., Palazzo, S., et al.: A convolutional-transformer model for ffr and ifr assessment from coronary angiography. *IEEE Transactions on Medical Imaging* **43**(8), 2866–2877 (2024)
 19. Mineo, R., Salanitri, F.P., Bellitto, G., De Filippo, O., D’Ascenzo, F., Palazzo, S., Spampinato, C.: Coronary artery stenosis assessment in x-ray angiography through spatio-temporal attention for non-invasive ffr and ifr estimation. In: *BIOSTEC* (1). pp. 305–312 (2024)

20. Mineo, R., Salanitri, F.P., Passarello, L., Sardella, A., Messina, L., Pennisi, M., Giordano, D., Palazzo, S., Spampinato, C.: Magnifying facial micro-movements for cognitive evaluation. In: 2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). pp. 1–5. IEEE (2024)
21. Mineo, R., Sorrenti, A., Proietto Salanitri, F.: Fedetr: A federated approach for stenosis detection in coronary angiography. In: International Conference on Image Analysis and Processing. pp. 189–200. Springer (2023)
22. Singh, P., Singh, S.: Chestx-transcribe: a multimodal transformer for automated radiology report generation from chest x-rays. *Frontiers in Digital Health* **7** (2025)
23. Tokushige, S.i., Matsumoto, H., Matsuda, S.i., Inomata-Terada, S., Kotsuki, N., et al.: Early detection of cognitive decline in alzheimer’s disease using eye tracking. *Frontiers in Aging Neuroscience* **15**, 1123456 (2023)
24. Wahul, R.M., Ambadekar, S., Dhanvijay, D.M., Dhanvijay, M.M., Dudhedia, M.A., et al.: Multimodal approaches and ai-driven innovations in dementia diagnosis: a systematic review. *Discover Artificial Intelligence* **5**(1), 96 (2025)
25. Xu, Y., Zhang, C., Pan, B., Yuan, Q., Zhang, X.: A portable and efficient dementia screening tool using eye tracking machine learning and virtual reality. *NPJ Digital Medicine* **7**(1), 219 (2024)
26. Yoonesi, S., Abedi Azar, R., Arab Bafrani, M., Yaghmayee, S., et al.: Facial expression deep learning algorithms in the detection of neurological disorders: a systematic review and meta-analysis. *BioMedical Engineering OnLine* **24**(1), 64 (2025)
27. Zhao, K., Zhang, K., Zhai, Y., Wang, D., Su, J.: Real-time sign language recognition based on video stream. *International Journal of Systems, Control and Communications* **12**(2), 158–174 (2021)